Layered Systems

Definition

A **layered system** is an operating-system structure in which the system is organized as a hierarchy of layers. Each layer is built on top of the layer below it and provides services to the layer above it.

1. Basic Idea

Main Concept

In the layered approach:

- the operating system is divided into several layers,
- each layer has its own function,
- each higher layer uses the services of the lower layer,
- this makes the system more organized and easier to understand.

Main Advantage

The layered design improves structure, clarity, and separation of responsibilities inside the operating system.

2. THE Operating System

Historical Example

The first important layered operating system was the **THE system**, built at the Technische Hogeschool Eindhoven in the Netherlands by E. W. Dijkstra and his students.

System Type

THE was a simple **batch system** built for the **Electrologica X8** computer.

Interesting Note

The machine had:

- 32K of 27-bit words,
- limited memory by modern standards,
- a design where careful organization was very important.

3. Six Layers of THE System

Layer Structure

The THE operating system was divided into **six layers**, numbered from 0 to 5.

Layer	Function
5	The operator
4	User programs
3	Input/output management
2	Operator-process communication
1	Memory and drum management
0	Processor allocation and multiprogramming

3.1 Simple Layer Diagram

Layer 5: The operator

Layer 4: User programs

Layer 3: I/O management

Layer 2: Operator-process communication

Layer 1: Memory and drum management

Layer 0: Processor allocation and multiprogramming

4. Layer 0

Processor Allocation

Layer 0 handled:

- allocation of the processor,
- switching between processes,
- response to interrupts and timer expiration.

Role

This layer provided the basic **multiprogramming** of the CPU. Above this layer, processes could behave as if each had its own processor.

5. Layer 1

Memory Management

Layer 1 managed:

- main memory,
- a 512K word drum,
- movement of process pages between them.

Main Idea

Above layer 1, a process did not need to worry about whether it was in main memory or on the drum. Layer 1 handled that automatically.

6. Layer 2

Operator Communication

Layer 2 handled communication between each process and the operator console.

Effect

Above this layer, each process effectively had its own operator console.

7. Layer 3

I/O Management

Layer 3 managed:

- I/O devices,
- buffering of input streams,
- buffering of output streams.

Abstraction

Above layer 3, processes worked with abstract I/O devices instead of dealing directly with real hardware devices and their peculiar behavior.

8. Layer 4

User Programs

Layer 4 contained the **user programs**.

Benefit

User programs at this level did not need to handle:

- processor management,
- memory management,
- operator console handling,
- or I/O device control.

9. Layer 5

Operator Layer

Layer 5 contained the **system operator process**.

10. Advantages of Layered Systems

Advantages

Layered systems provide:

- better organization,
- clearer structure,
- easier understanding,
- separation of responsibilities,
- simpler system design.

11. MULTICS and Ring Structure

Generalization

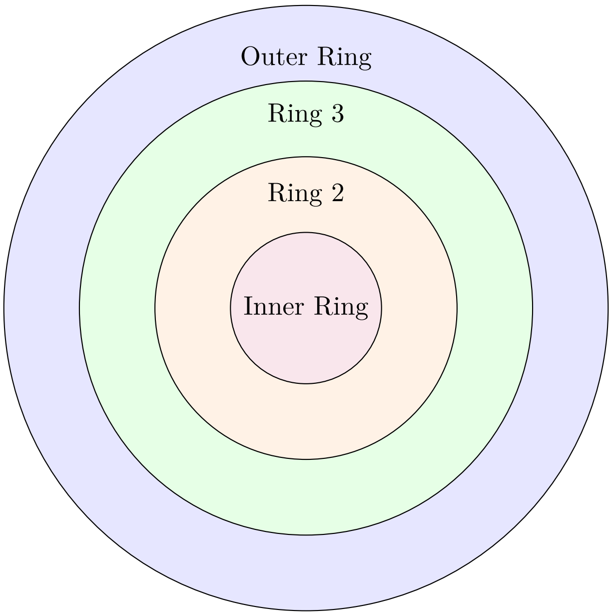
A further generalization of layering appeared in the **MULTICS** system. Instead of layers, MULTICS used a system of **concentric rings**.

Ring Idea

In MULTICS:

- inner rings were more privileged,
- outer rings were less privileged,
- outer-ring procedures needed special permission to call inner-ring procedures.

11.1 Ring Model Diagram



System Call Style

If a procedure in an outer ring wanted to call a procedure in an inner ring, it had to perform the equivalent of a **system call**. Its parameters were carefully checked before access was allowed.

12. Protection in MULTICS

Hardware Enforcement

In MULTICS, the ring mechanism was not just a design idea. It was actually enforced by the **hardware** at run time.

Memory Protection

Although the whole operating system was in the address space of each user process, the hardware could protect individual procedures or memory segments against:

- reading,
- writing,
- executing.

13. THE vs MULTICS

Important Difference

The layering of THE was mainly a **design aid**, because all parts were eventually linked into one executable program.

MULTICS Advantage

In MULTICS, the ring mechanism existed at **run time** and was enforced by hardware, making it stronger and more secure than simple design-time layering.

14. Use of Rings for User Subsystems

Extension of Ring Concept

The ring model could also be used to structure user subsystems, not just the operating system.

Example

A professor could run a grading program in ring n, while student programs ran in ring $n+1$. This would prevent student programs from changing their grades.

15. Summary Points

Quick Review

A layered system:

- organizes the operating system into levels,
- lets each layer depend on the one below,
- improves structure and abstraction,
- was first demonstrated in the THE system,
- was further generalized by the ring model of MULTICS.